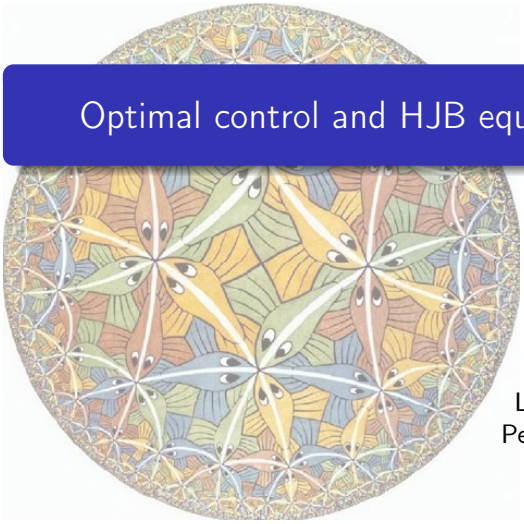


Optimal control and HJB equations in non-positively curved spaces

Averil Aussedat
Joint work with Othmane Jerhaoui and Hasnaa Zidani

November 12, 2026
LAMPS Seminar
Perpignan

LMI - EA 3226
Università di Pisa

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- get a formulation of ODEs that would allow for Cauchy-Lipschitz theorems;
- see whether the associated control problems can be treated with the (previously existing) theory of Hamilton-Jacobi equations in these spaces.

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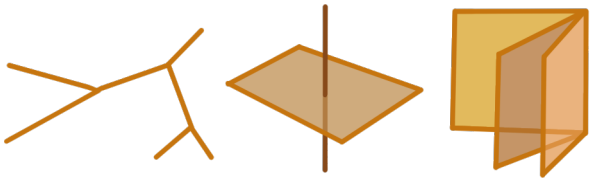
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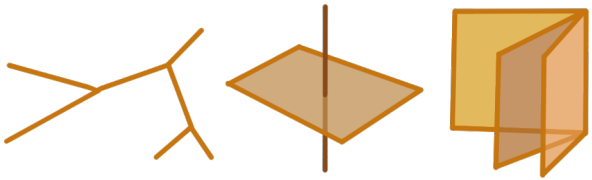
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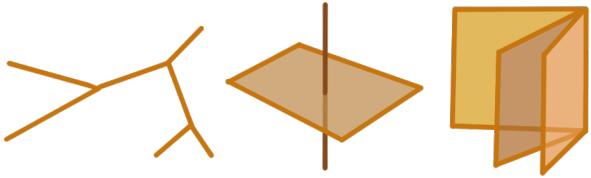


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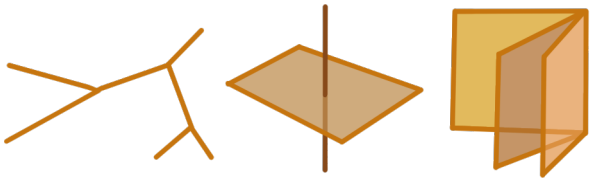
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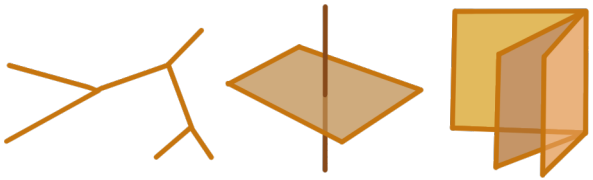
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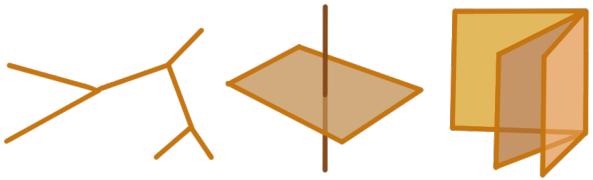
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Observation of O. Jerhaoui: gluings are instances of *CAT(0) spaces*.

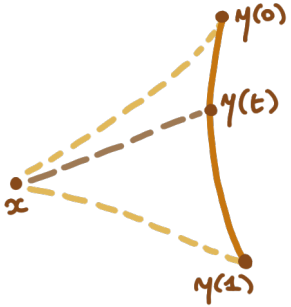
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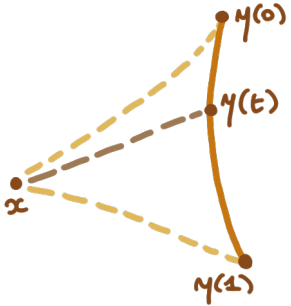
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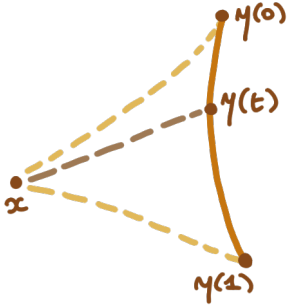
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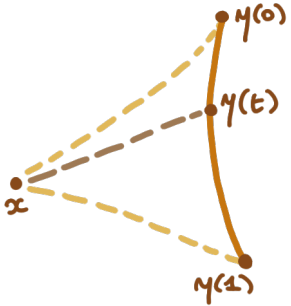
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- Example which is *not* so: the sphere. ⚠ At best local models (for traffic networks, for instance).

Canonical construction of tangent cones

Manipulating the squared distance inequality, we get that for two geodesics γ, γ' issued from the same x,

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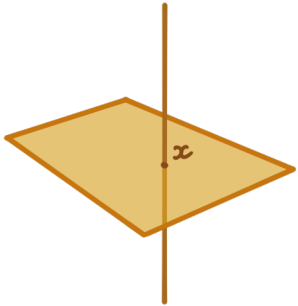
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- Provides curves along which one can differentiate.

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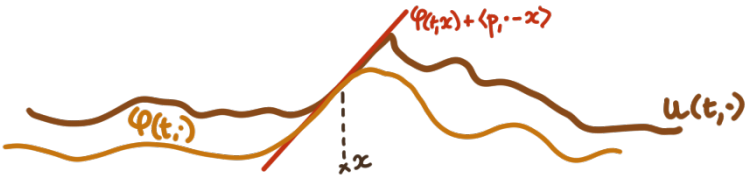
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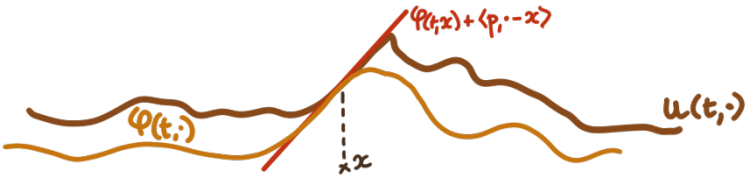
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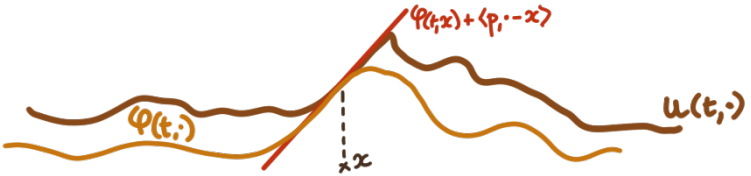
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- Smoothness of test functions gives stability. What if Ω does not support smooth functions?

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Focus for the rest of the talk: continuous equations/Hamiltonians, setting of CAT(0) spaces.

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Question now: can we apply it to control problems (which were the motivation...) ?

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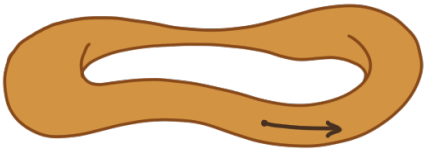
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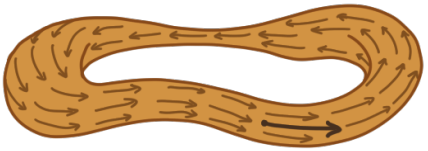


Definition of ODEs

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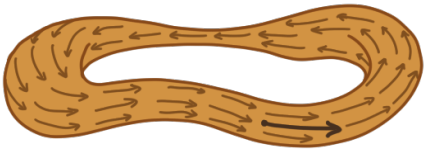


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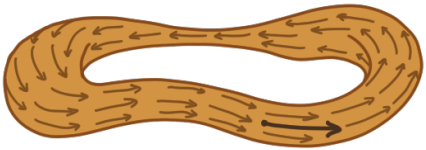


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- Extensive theory of existence, well-posedness, estimates par to Cauchy-Lipschitz, Filippov theorems, invariance and viability, application to control of sets...

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Possible (and fruitful) to consider other families than $(\alpha, x_0) \mapsto \alpha d(\cdot, x_0)$.

Well-posedness of controlled systems

For the sequel, U is a compact set of controls.

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Here $\theta := f(y_t, u(t))$ is a transition, hence a semigroup, and $\theta(h, y_t)$ is the associated curve starting from y_t and evolving until time h.

Control problem

Now that we have controlled systems, we can consider *control problems*, for instance

$$\begin{cases} \text{Minimize} \quad \mathfrak{J}(y_T^{x,u}) \text{ over controls } u \in L^0([0,T];U) \\ \text{where } (y_t^{x,u})_t \text{ solves the previous system with control } u(\cdot). \end{cases} \tag{CP}$$

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Using the results of mutational analysis, the Dynamical Programming Principle and the local Lipschitz character of V follow exactly as in the classical case. What about HJ?

Hamilton-Jacobi-Bellman equation

If φ is directionally differentiable everywhere, and $\theta \in \Theta$, let $L_\theta \varphi$ be the directional derivative

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To a dynamic $f : \Omega \times U \to \Theta$, associate a Hamiltonian

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Theorem – [AZ25] The value function is the unique viscosity solution of the HJ equation

$$-\partial_t u(t,x) + H(x, D_x u(t,x)) = 0 \quad \text{in } (0,T) \times \Omega, \qquad u(T, \cdot) = \mathfrak{J}.$$

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Not all control problems are well-posed (in the variable u); consider

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👉 Need to convexify the set of admissible dynamics to obtain existence.
Here, one would replace $\{-1,1\}$ by $[-1,1]$, and the control $u(\cdot) \equiv 0$ is optimal.

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In fact, (c) can be used to obtain existence, and implies both (a) and (b).

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$$E_\omega \coloneqq \int_{u \in U} E_{f(x,u)} d\omega(u).$$

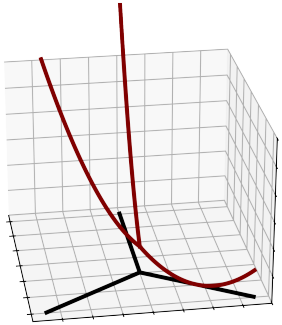
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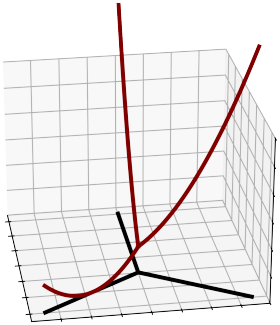
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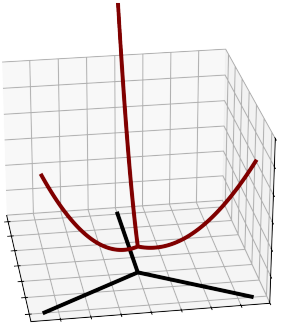
$$E_\omega := \int_{u \in U} E_{f(x,u)} d\omega(u).$$

The convexification $F : \Omega \times \mathscr{P}(U) \to \Theta_{\mathsf{conv}}$ is defined by $F(x, \omega) := \text{Gradient Flow}(E_\omega)$.

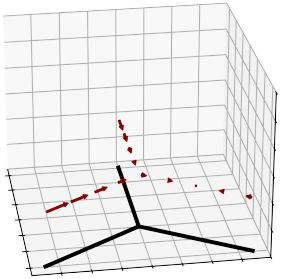
Example: tripod network

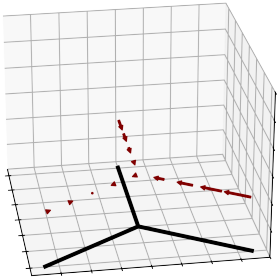


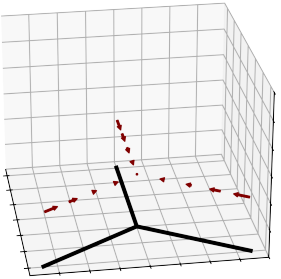




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When looking at a given point $x \in \Omega$, under some natural assumptions over Θ, one has that

the metric gradient of the convexification is the barycenter of the metric gradients

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Namely, compute $\hat{V}_n(\hat{x})$ for each point $\hat{x}$ of a grid, with

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PDE-based schemes have been studied for gluings by Costeseque, Lebacque and Monneau 2015, Guerand and Koumaiha 2019, Morfe 2020, and Bokanowski, Jerhaoui and Zidani 2025.

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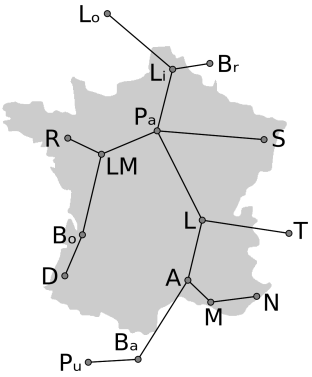
Proposition Under (A1) and Lipschitz assumption on f, there holds

$$\left|\hat{V}_n(\hat{x}) - V(t_n, \hat{x})\right| \leqslant C_{T,\mathsf{Lip}(V)} \left(\frac{\Delta x}{\Delta t} + \Delta t\right).$$

Hence the scheme is convergent with order 1/2 under the inverse CFL condition $\Delta t = \sqrt{\Delta x}$.

Example: the SNCF network

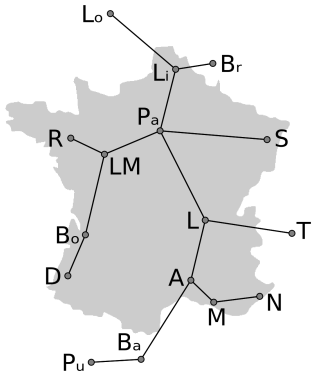
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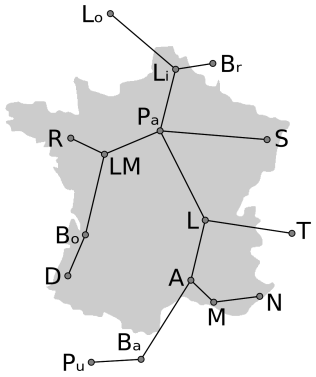


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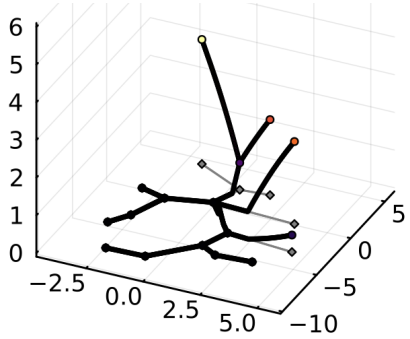


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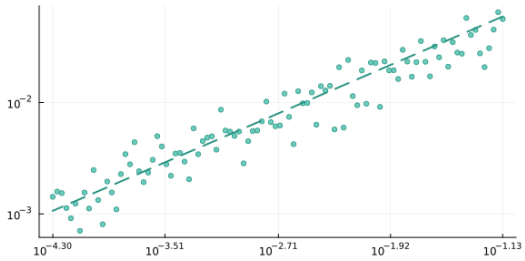


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Relative error on the value function w.r.t. Δx.
The slope is 0.55.

Example: a robust problem (1/2)

Consider a problem in dimension $d = 2$, with G sym. def. pos:

$$\begin{cases} \text{Minimize } \max_{x \in \overline{\mathscr{B}}(0,1)} \langle G y_T^{x,u}, y_T^{x,u} \rangle \text{ over controls } u \in L^1(0,T;U), \\ \text{where } y_0^{x,u} = x \text{ and } \dot{y}_t^{x,u} = A(u(t)) y_t^{x,u}. \end{cases}$$

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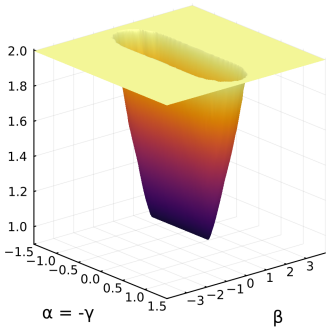
☛ Classical control problem with cost $\mathfrak{J} = \lambda_{\max}$, on trajectories of sdp matrices satisfying the ODE

$$G^u(0) = G, \qquad \dot{G}^u(t) = A^*(u(t)) G^u(t) + G^u(t) A(u(t)).$$

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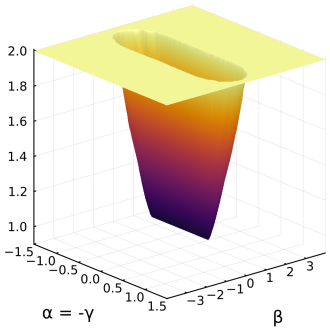
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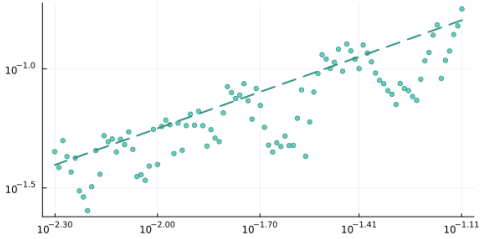


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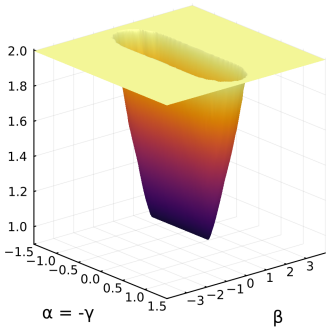


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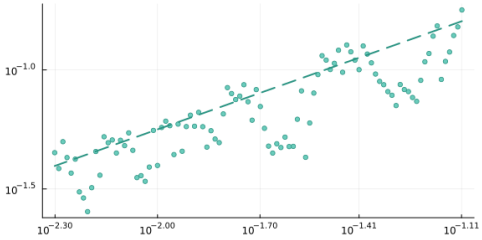


Relative error on the value function w.r.t. Δx. The slope is 0.627.

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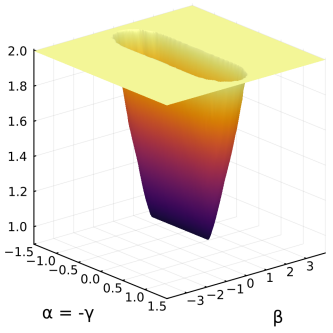
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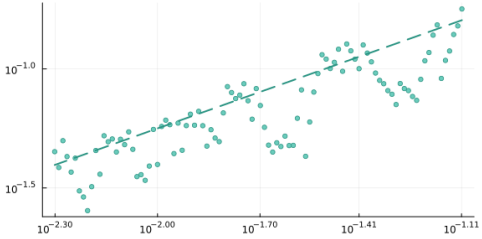
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- Quite heavy numerically: computation of geodesics, meshes, semi-Lagrangian itself...
- The projection on the grid should be replaced by a higher-order method (classically, interpolation): not trivial in general CAT(0) spaces.

Open directions

Interesting points include:

- Original aim of this work: populations over networks.

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Thank you for you attention.

Talk based on the preprint "A Cauchy-Lipschitz setting for control problems in complete CAT(0) spaces", with H. Zidani.
Details of numerical examples are given in my PhD manuscript (available on HAL theses),
check also Othmane's for details of theoretical results on HJ equations.

Julia code available on `github.com/averil-aussedat/FLagHada.jl`.